Hongxin Zhang

University of Massachusetts, Amherst
Email: hongxinzhang@umass.edu
Tel: 4132759957
Homepage: https://icefoxzhx.github.io/
Google Scholar: 2,300+ citations, h-index 13

Education

University of Massachusetts Amherst, Amherst, MA, USA — Sep. 2023 - May 2027 (expected)
Ph.D. in Computer Science — *Advisor: Chuang Gan*

University of Massachusetts Amherst, Amherst, MA, USA — Sep. 2023 - May 2025
Master of Science in Computer Science — *Advisor: Chuang Gan*

Shanghai Jiao Tong University, Shanghai, China — Sep. 2019 - June 2023
B.E. in Computer Science, ACM Honors Class — *Advisor: Yong Yu, Diyi Yang*

Publications

1. Chunru Lin*, **Hongxin Zhang***, Fenghao Yu, Zhehuan Chen, Thomas L. Griffiths, Yejin Choi, David Held, Chuang Gan. "RoboWits: Unexpected Challenges for Robotic Creative Problem Solving" *Neural Information Processing Systems (NeurIPS) Evaluations & Datasets Track*, 2026
2. Qinhong Zhou*, Riley Zong*, Chunru Lin, Shivraj Singh Bhatti, **Hongxin Zhang**, Diyi Hu, Lin Qiu, Hanqing Zeng, Chuang Gan. "HARP-Bench: Benchmarking Robot Manipulation Around Active Humans" *Conference on Robot Learning (CoRL)*, 2026
3. Xiangye Lin*, **Hongxin Zhang***, Ruxi Deng, Qinhong Zhou, Chuang Gan. "Sentinel: Embodied Cooperative Spatial Reasoning and Planning" *European Conference on Computer Vision (ECCV)*, 2026
4. Qinhong Zhou*, **Hongxin Zhang***, Xiangye Lin*, Zheyuan Zhang*, Yutian Chen, Wenjun Liu, Zunzhe Zhang, Sunli Chen, Lixing Fang, Qiushi Lyu, Xinyu Sun, Jincheng Yang, Zeyuan Wang, Bao Chi Dang, Zhehuan Chen, Daksha Ladia, Jiageng Liu, Chuang Gan. "Virtual Community: An Open World for Humans, Robots, and Society" *International Conference on Learning Representations (ICLR)*, 2026.
5. **Hongxin Zhang***, Zheyuan Zhang*, Zeyuan Wang*, Zunzhe Zhang, Lixing Fang, Qinhong Zhou, Chuang Gan. "Ella: Embodied Social Agents with Lifelong Memory" *International Conference on Machine Learning (ICML) MAS Workshop*, 2025
6. Haoyu Zhen*, Qiao Sun*, **Hongxin Zhang**, Junyan Li, Siyuan Zhou, Yilun Du, Chuang Gan. "TesserAct: Learning 4D Embodied World Models" *International Conference on Computer Vision (ICCV)*, 2025
7. Yuncong Yang*, Han Yang*, Jiachen Zhou, Peihao Chen, **Hongxin Zhang**, Yilun Du, Chuang Gan. "3D-Mem: 3D Scene Memory for Embodied Exploration and Reasoning" *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025
8. **Hongxin Zhang***, Zeyuan Wang*, Qiushi Lyu*, Zheyuan Zhang, Sunli Chen, Tianmin Shu, Yilun Du, Behzad Dariush, Kwonjoon Lee, Chuang Gan. "COMBO: Compositional World Models for Embodied Multi-Agent Cooperation" *International Conference on Learning Representations (ICLR)*, 2025.
9. Weihua Du*, Qiushi Lyu*, Jiaming Shan, Zhenting Qi, **Hongxin Zhang**, Sunli Chen, Andi Peng, Tianmin Shu, Kwonjoon Lee, Behzad Dariush, Chuang Gan. "Constrained Human-AI Cooperation: An Inclusive Embodied Social Intelligence Challenge" *Neural Information Processing Systems (NeurIPS)*, 2024.
10. **Hongxin Zhang***, Weihua Du*, Jiaming Shan, Qinhong Zhou, Yilun Du, Joshua B. Tenenbaum, Tianmin Shu, Chuang Gan. "Building Cooperative Embodied Agents Modularly with Large Language Models" *International Conference on Learning Representations (ICLR)*, 2024.
11. Qinhong Zhou, Sunli Chen, Yisong Wang, Haozhe Xu, Weihua Du, **Hongxin Zhang**, Yilun Du, Joshua B. Tenenbaum, Chuang Gan. "HAZARD Challenge: Embodied Decision Making in Dynamically Changing Environments" *International Conference on Learning Representations (ICLR)*, 2024.
12. Zhiqing Sun, Yikang Shen, **Hongxin Zhang**, Qinhong Zhou, Zhenfang Chen, David Cox, Yiming Yang, Chuang Gan. "SALMON: Self-Alignment with Principle-Following Reward Models" *International Conference on Learning Representations (ICLR)*, 2024.
13. Zhiqing Sun, Yikang Shen, Qinhong Zhou, **Hongxin Zhang**, Zhenfang Chen, David Cox, Yiming Yang, Chuang Gan. "Principle-Driven Self-Alignment of Language Models from Scratch with Minimal Human Supervision" *Neural Information Processing Systems (NeurIPS)*, 2023.
14. Omar Shaikh, **Hongxin Zhang**, William Held, Michael Bernstein, Diyi Yang. "On Second Thought, Let's Not Think Step by Step! Bias and Toxicity in Zero-Shot Reasoning" *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.
15. Bolin Lai*, **Hongxin Zhang***, Miao Liu*, Aryan Pariani*, Fiona Ryan, Wenqi Jia, Shirley Anugrah Hayati, James M. Rehg, Diyi Yang. "Werewolf Among Us: A Multimodal Dataset for Modeling Persuasion Behaviors in Social Deduction Games" *Findings of the Association for Computational Linguistics: ACL*, 2023.
16. Albert Lu*, **Hongxin Zhang***, Yanzhe Zhang, Xuezhi Wang, Diyi Yang. "Bounding the Capabilities of Large Language Models in Open Text Generation with Prompt Constraints" *Findings of the Association for Computational Linguistics: EACL*, 2023.

17. **Hongxin Zhang**, Yanzhe Zhang, Ruiyi Zhang, Diyi Yang. "Robustness of Demonstration-based Learning Under Limited Data Scenario" *Empirical Methods in Natural Language Processing (EMNLP)*, 2022.

PREPRINTS

1. **Hongxin Zhang**, Chunru Lin, Junyan Li, Zhou Xian, Tsun-Hsuan Wang, Chuang Gan. "GS-Agent: Creating 4D Physical Worlds With Generative Simulation"
2. **Hongxin Zhang***, Chunru Lin*, Tsun-Hsuan Wang, Zhenjia Xu, Chuang Gan. "Self-Adaptive VLA for Robust Robot Deployment"

RESEARCH EXPERIENCE

University of Massachusetts Amherst — Amherst, Massachusetts

Graduate Research Assistant, advised by Prof. Chuang Gan — Sep. 2023 - Present

- **RoboWits: Unexpected Challenges for Robotic Creative Problem Solving**
  Chunru Lin*, **Hongxin Zhang***, Fenghao Yu, Zhehuan Chen, Thomas L. Griffiths, Yejin Choi, David Held, Chuang Gan
  NeurIPS 2026 Evaluations & Datasets Track
  - Introduced a bi-manual robotic benchmark that systematically evaluates cognitive reasoning, creative tool use, and robustness to unexpected conditions.
  - Proposed a multi-agent pipeline automating seed task generation and verification, metric generation, scene generation, and task mutation, curating 30 seed tasks and 170 mutated tasks across geometry, material, and assembly-based reasoning.
  - Benchmarked popular robot policies, pre-trained VLAs, and oracle-state planners, revealing that fine-tuned VLAs show preliminary success on seed tasks but struggle on mutated tasks that require reasoning and strategy adaptation.
- **COMBO: Compositional World Models for Embodied Multi-Agent Cooperation**
  Hongxin Zhang*, Zeyuan Wang*, Qiushi Lyu*, Zheyuan Zhang, Sunli Chen, Tianmin Shu, Yilun Du, Behzad Dariush, Kwonjoon Lee, Chuang Gan
  ICLR 2025 Poster
  - Proposed to learn a compositional world model for multi-agent cooperation by factorizing joint actions of an arbitrary number of agents and compositionally generating the video to enable accurate simulation of multiple sets of actions on the world state.
  - Introduced a Compositional World Model-based embodied multi-agent planning framework to empower the agents to imagine how different actions may affect the world with other agents in the long run and plan more cooperatively.
  - Trained a 200M video diffusion model from scratch, another two smaller diffusion models for inpainting and super-resolution, and LoRA fine-tuned LLaVA-7B as planning sub-modules.
- **Building Cooperative Embodied Agents Modularly with Large Language Models**
  Hongxin Zhang*, Weihua Du*, Jiaming Shan, Qinhong Zhou, Yilun Du, Joshua B. Tenenbaum, Tianmin Shu, Chuang Gan
  ICLR 2024 Poster
  - Conducted the first systematic study on LLMs' capacity for planning and communication in embodied multi-agent cooperation.
  - Introduced a novel framework that utilizes LLMs to build cooperative embodied agents, integrating perception, memory, planning, communication, and motion, surpassing strong planning-based methods.
  - Conducted a user study to evaluate whether LLMs can enable effective and trustworthy human-AI cooperation.

Stanford University — Stanford, California

Visiting Student Researcher, advised by Prof. Diyi Yang — Sep. 2022 - Jan. 2023

- **On Second Thought, Let's Not Think Step by Step! Bias and Toxicity in Zero-Shot Reasoning**
  Omar Shaikh, **Hongxin Zhang**, William Held, Michael Bernstein, Diyi Yang **ACL 2023**
  - Performed a controlled evaluation of zero-shot CoT across two sensitive domains: harmful questions & stereotype benchmarks.
  - Found that using zero-shot CoT reasoning in a prompt can significantly increase a model's likelihood of producing undesirable output.
- **Werewolf Among Us: A Multimodal Dataset for Modeling Persuasion Behaviors in Social Deduction Games**
  Bolin Lai*, **Hongxin Zhang***, Miao Liu*, Aryan Pariani*, Fiona Ryan, Wenqi Jia, Shirley Anugrah Hayati, James M. Rehg, Diyi Yang. **ACL 2023**
  - Presented the first multimodal dataset for persuasion modeling, which is collected in naturalistic social game scenarios with intensive in-person conversations with multiple players.
  - Conducted comprehensive experiments to show the importance of context and visual signals for persuasion strategy prediction.
  - Provided additional experiment results to investigate model generalization on the persuasion modeling task, and discussed how persuasion strategy influences the game voting outcome.
- **Robustness of Demonstration-based Learning Under Limited Data Scenario**
  Hongxin Zhang, Yanzhe Zhang, Ruiyi Zhang, Diyi Yang. **EMNLP 2022**
  - Investigated the robustness of demonstration-based learning by designing pathological demonstrations.
  - Experiments show demonstrations composed of random tokens still make the model a better few-shot learner; the length of random demonstrations and the relevance of random tokens are the main factors affecting the performance; demonstrations increase the confidence of model predictions on captured superficial patterns.

Industrial Experiences

Genesis AI San Carlos, California

Research Scientist Intern Jun. 2025 - Present

- **GS-Agent: Creating 4D Physical Worlds With Generative Simulation**
  - Built a multi-agent system that creates 4D worlds from natural language, producing **physically plausible interactions** among liquids, deformable objects, and rigid bodies, with **cinematic camera and lighting controls**.
  - Designed harnesses that expose the physics engine codebase as agent tools, and optimized the agent loop with multi-modal feedback and coding tools.
- **Self-Adaptive VLA for Robust Robot Deployment**
  - Developed a post-training method that lets Vision-Language-Action policies learn from sub-optimal rollouts to adapt to deployment hardware shifts such as actuation bias and joint encoder offsets.
  - Deployed on real robots across four precision-critical tasks: 6-DoF Piper/Piper-X arms with parallel grippers, and bi-manual 7-DoF Marvin arms with 20-DoF Wuji dexterous hands.

Adobe San Jose, California

Research Scientist/Engineer Intern @ Document Intelligence Lab Jun. 2024 - Aug. 2024

- **Proactive Document Agents**
  - Designed harnesses that let LLM agents interact with documents more naturally and efficiently.
  - Collected a dataset of user interactions across diverse documents, including forms, contracts, and brochures.
  - Built document agents that infer user intent from interaction history and offer to automate repetitive edits.

TEACHING EXPERIENCE

Teaching Assistant: Great Ideas in Computer Science Fall 2020

Student Instructor: Principle and Practice of Computer Algorithms Summer 2021

Teaching Assistant: Machine Learning Spring 2022

Teaching Assistant: COMPSCI-603 Robotics Spring 2025

SKILLS AND INTERESTS

Programming: Python / C++ / Java / Go / Verilog

Language: Mandarin (native), English (TOEFL 109/120), Latin (Beginner)

Interests: Literature, Movie, Billiards

Honors and Awards

Shanghai Excellent Graduate (Awarded for overall performance in undergraduate career) 2023

Zhiyuan Outstanding Student Scholarship (Highest award for undergraduate in SJTU) 2023

Foresight-Sequoia Scholarship (**5** winners at Zhiyuan College) 2022

Academic Excellence Scholarship 2020, 2021, 2022

35th China's National Olympiad in Informatics (NOI) Silver Medal 2018